80386DX Question Bank

Topics: Salient features of the 80386DX, architecture and signal description, register organization and addressing modes, data types, real address mode, protected mode, segmentation and paging.

Part A: 2-Mark Questions

1. List any four salient features of the 80386DX.
2. Why is the 80386DX called a 32-bit processor? State its address-bus and data-bus widths.
3. Name the six functional units of the 80386 architecture.
4. What is the function of the prefetch unit and the instruction queue?
5. Why are address lines A0 and A1 not available in the 80386? How are they replaced?
6. List the eight 32-bit general-purpose registers of the 80386.
7. What is a scaled-index addressing mode? Give an example.
8. Differentiate between real address mode and protected mode.
9. What is a segment descriptor? State its size.
10. What are the sizes of a page and of a page table entry in the 80386 paging unit? Name the registers CR2 and CR3 used for.

Answers to Part A

1. 32-bit data and address bus, 4 GB physical memory, 64 TB virtual memory, on-chip segmentation and paging, three operating modes (real, protected, virtual 8086), pipelined architecture, 132-pin PGA package.
2. Its internal registers and ALU are 32 bits wide, and it has a 32-bit address bus (A31–A2 with BE0#–BE3#) and a 32-bit data bus (D31–D0).
3. Bus interface unit, code prefetch unit, instruction decode unit, execution unit, segmentation unit, paging unit.
4. It fetches instructions in advance from memory into a 16-byte queue, whenever the bus is free, so the execution unit does not wait for fetches.
5. The bus is 32 bits wide, so addresses are doubleword-aligned. Four byte-enable signals (BE0#–BE3#) select the bytes within the doubleword instead.
6. EAX, EBX, ECX, EDX, ESI, EDI, EBP, ESP.

7. The index register is multiplied by a scale factor of 1, 2, 4 or 8. Example: `MOV EAX, [EBX + ESI*4 + 10H]`.
8. Real mode is an 8086-style mode with a 1 MB address space, 64 KB segments and no protection. Protected mode provides 4 GB physical memory, descriptor-based segmentation, paging, multitasking and privilege protection.
9. It is an 8-byte entry in the GDT or LDT that holds a segment's base address, limit and access rights.
10. A page is 4 KB and a page table entry is 4 bytes. CR2 holds the linear address that caused a page fault. CR3 holds the physical base address of the page directory.

Part B: 20 Objective Questions (MCQs)

1. The 80386DX has a data bus width of: a) 16 bits b) 32 bits c) 64 bits d) 24 bits
2. The physical address space of the 80386DX is: a) 1 MB b) 16 MB c) 4 GB d) 64 TB
3. The virtual address space of the 80386 is: a) 4 GB b) 16 GB c) 64 TB d) 1 TB
4. The 80386DX is housed in a: a) 40-pin DIP b) 68-pin PLCC c) 132-pin PGA d) 64-pin QFP
5. The size of the code prefetch queue is: a) 4 bytes b) 8 bytes c) 16 bytes d) 32 bytes
6. Which signal indicates a valid bus cycle address? a) READY# b) ADS# c) NA# d) HOLD
7. The 80386 coprocessor is: a) 8087 b) 80287 c) 80387 d) 80487
8. The number of segment registers in the 80386 is: a) 4 b) 5 c) 6 d) 8
9. The new segment registers added in the 80386 over the 8086 are: a) FS and GS b) ES and FS c) GS and SS d) DS and ES
10. The 32-bit flag register of the 80386 is: a) FLAGS b) EFLAGS c) CR0 d) IPR
11. Maximum scale factor in the scaled-index addressing mode is: a) 2 b) 4 c) 8 d) 16
12. In real mode the 80386 can address a maximum of: a) 64 KB b) 1 MB c) 16 MB d) 4 GB
13. Maximum segment size in real mode is: a) 4 KB b) 64 KB c) 1 MB d) 4 GB
14. The bit in CR0 that switches the processor into protected mode is: a) PG b) PE c) TS d) EM
15. The size of a segment descriptor is: a) 2 bytes b) 4 bytes c) 8 bytes d) 16 bytes
16. A segment selector contains an index field of: a) 8 bits b) 12 bits c) 13 bits d) 16 bits
17. The number of privilege levels in protected mode is: a) 2 b) 3 c) 4 d) 8
18. The paging unit converts a: a) logical address to linear address b) linear address to physical address c) physical to logical address d) selector to descriptor
19. The page size in the 80386 is: a) 1 KB b) 2 KB c) 4 KB d) 64 KB

20. The 80386 TLB (translation lookaside buffer) holds: a) 8 entries b) 16 entries c) 32 entries d) 64 entries

Answer Key for Part B

1-b, 2-c, 3-c, 4-c, 5-c, 6-b, 7-c, 8-c, 9-a, 10-b, 11-c, 12-b, 13-b, 14-b, 15-c, 16-c, 17-c, 18-b, 19-c, 20-c

Part C: 20 Fill in the Blanks

1. The 80386DX is fabricated using _____ technology (CHMOS III).
2. The 80386DX has a _____-bit address bus.
3. The three operating modes of the 80386 are real mode, protected mode and _____ mode.
4. The signal that inserts wait states when the memory or I/O is slow is _____.
5. The byte-enable signals of the 80386 are _____ to _____.
6. The signal that requests the bus by another bus master is _____, and it is acknowledged by _____.
7. The signal _____ distinguishes between memory and I/O cycles (M/IO#).
8. The input that supports a 16-bit data bus width dynamically is _____.
9. The 80386 instruction pointer is called _____.
10. The system address registers of the 80386 are GDTR, IDTR, LDTR and _____.
11. The control registers of the 80386 are CR0, CR1, CR2 and _____.
12. The debug registers of the 80386 are numbered DR0 to _____.
13. In real mode, the physical address is formed by shifting the segment base left by _____ bits and adding the offset.
14. After reset, the 80386 begins execution at physical address _____.
15. A 48-bit pointer in the 80386 consists of a 16-bit selector and a _____-bit offset.
16. A data type consisting of 32 bits is called a _____.
17. The descriptor table that is common to all tasks is the _____.
18. The _____ bit (bit 31) of CR0 enables paging.
19. In a linear address, the upper _____ bits select the page directory entry, the next 10 bits select the page table entry, and the lower 12 bits give the offset.
20. The register that holds the physical base address of the page directory is _____.

Answer Key for Part C

1. CHMOS
2. 32
3. Virtual 8086
4. READY#
5. BE0# to BE3#
6. HOLD; HLDA
7. M/IO#
8. BS16#
9. EIP
10. TR (Task Register)
11. CR3
12. DR7
13. 4
14. FFFFFFF0H
15. 32
16. Doubleword
17. GDT (Global Descriptor Table)
18. PG
19. 10
20. CR3